# Expanded Resolved Candidate-Window Atlas

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**Abstract**

The frozen RH-244/RH-248 shell class was exhausted, so RH-251 required a genuinely expanded resolved window before another anchor test. We double the RH-222 candidate margin from 16 to 32 at all 32 noise/channel endpoints [1, 2]. The expanded eigensolver returns 16 additional bulk roots at every endpoint, and the archived roots match with maximum discrepancy $7.41 \times 10^{-9}$. A fixed-count window is not always shell-complete: 11 endpoints terminate in a split conjugate pair. After removing those boundary fragments, the complete expanded ranks lie between 33 and 64. This paper supplies the finite spectral input for RH-255; it does not yet test the deterministic anchor.

## 1 Expansion protocol

The RH-222 atlas uses noise scales [1]

$$0.04, 0.032, 0.025, 0.02, 0.016, 0.0125, 0.01, 0.008,$$
$$0.00625, 0.005, 0.004, 0.0032, 0.0025, 0.002, 0.0016, 0.00125.$$

and two channels (fine and Haar-coarse). At target rank $k$, the old resolution requested $k + 16$ leading eigenvalues; the present audit requests $k + 32$. The same deterministic starting vector, tolerance, and Hardy scaling are used, so the comparison is a controlled enlargement rather than a change of operator.

**Proposition 1.1** (Finite expanded-window stability)**.** *At each of the 32 endpoints the expanded computation resolves 16 more bulk roots than the archived window. Matching every old root to a distinct new root gives maximum error*

$$7.4054691027 \times 10^{-9},$$

*while Perron and parity discrepancies are below* $10^{-14}$*.*

*Proof.* The experiment constructs the same sparse folded-Gaussian matrix and calls the same deterministic eigensolver with the larger candidate count. A rectangular absolute-difference cost matrix is solved by a one-to-one assignment, and the reported maximum is the largest matched distance. The two distinguished roots are compared directly with their archived values. □

## 2 Conjugate-shell boundary

Because the matrices are real, a legitimate shell selector must use real roots or complete conjugate pairs. The final root of a finite eigensolver window can be one member of a pair. The audit therefore partitions the expanded roots by the RH-222 shell rule and reports both raw and shell-complete ranks.

| quantity | value |
|---|---|
| endpoints | 32 |
| raw expanded rank range | 34–64 |
| new raw roots per endpoint | 16 |
| shell-complete rank range | 33–64 |
| shell-complete endpoints | 21/32 |
| split-pair endpoints | 11/32 |
| maximum discarded boundary roots | 1 |
| maximum old/new matching error | $7.41 \times 10^{-9}$ |

The split-pair count is a boundary artifact of fixed-count resolution, not a claim that the underlying operator lacks conjugate symmetry. RH-255 uses only the complete shells and records the discarded fragments explicitly.

## 3 Scope and next result

This is a finite expanded candidate window, not an interval-certified spectral set and not a uniform small-noise theorem. No anchor has yet been shown to be reachable, and the old frozen obstruction is not silently reused. The next paper must perform an anchored zonotope/cone audit on the complete expanded shells. Gates A–E remain false/open; no Hilbert–Polya operator, zeta-divisor identification, Riemann-zero identification, or RH implication is asserted.

## References

[1] Bin Wang. Rank-growing conjugate-cloud atlas. *Preprint*, 2026. RH-222 preprint.

[2] Bin Wang. Anchored shell-zonotope reachability obstruction. *Preprint*, 2026. RH-248 preprint.